Parser Generators

Concepts of Programming Languages

CAS CS 320

Outline

- Extend our BNF syntax to be a bit more convenient
- Introduce **parser generators**
- Discuss **lexical analysis**
- Demo **Menhir,** the parser generator for this course

Recap

Recall: Example

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abstractions (non-terminal symbols)

tokens (terminal symbols)

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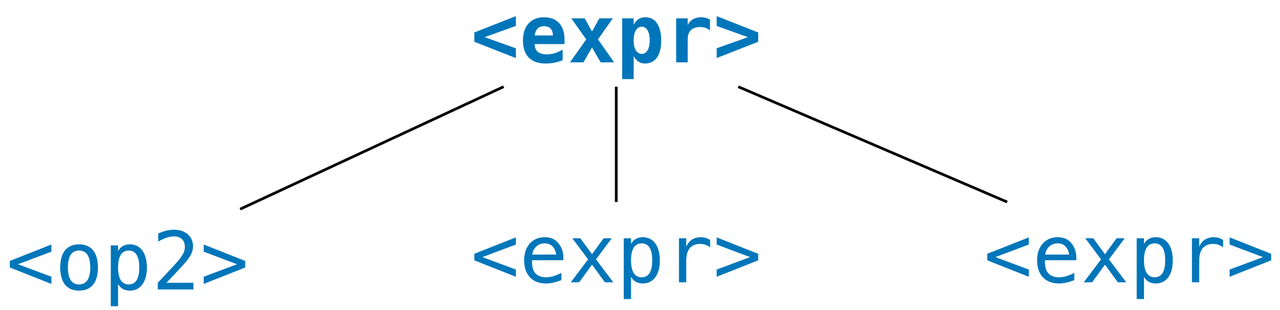

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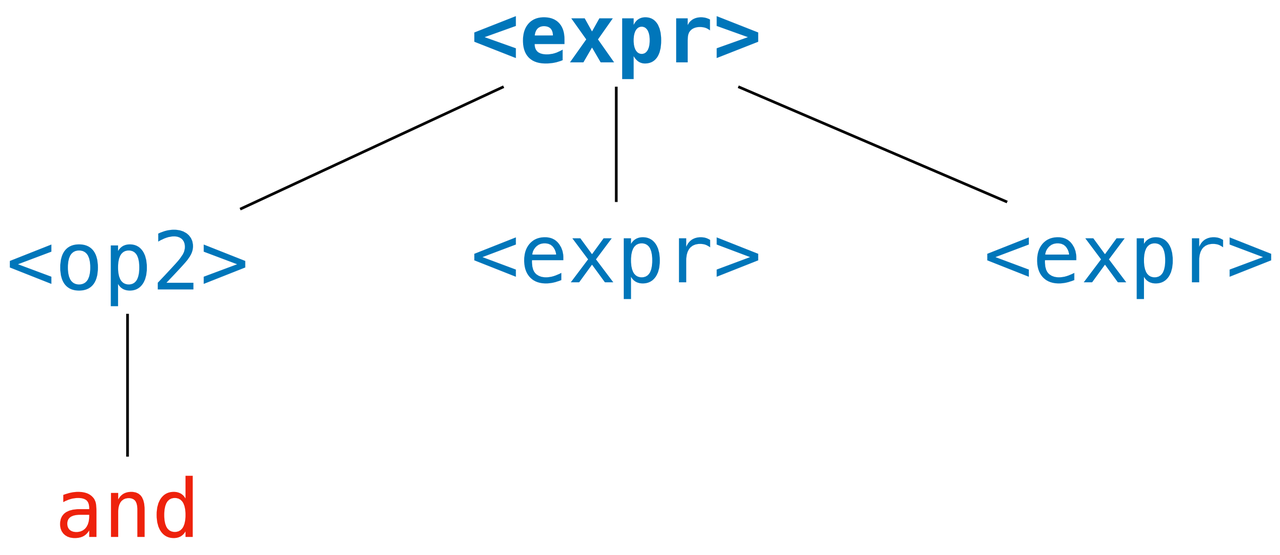

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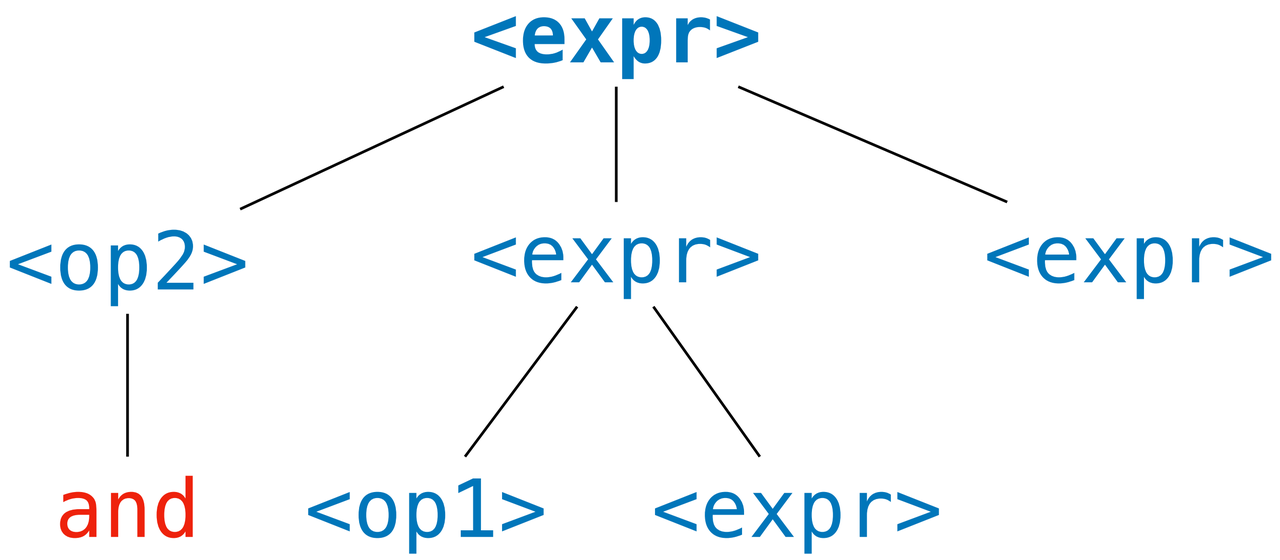

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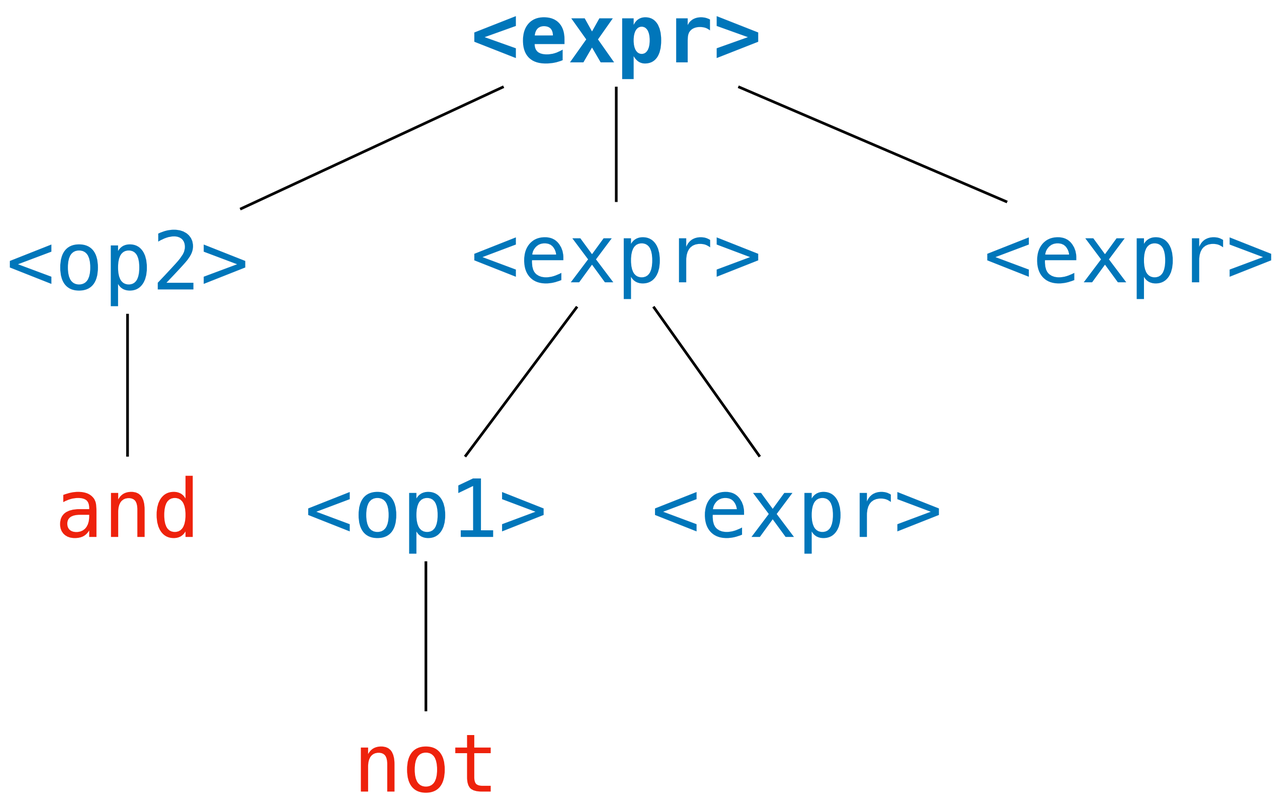

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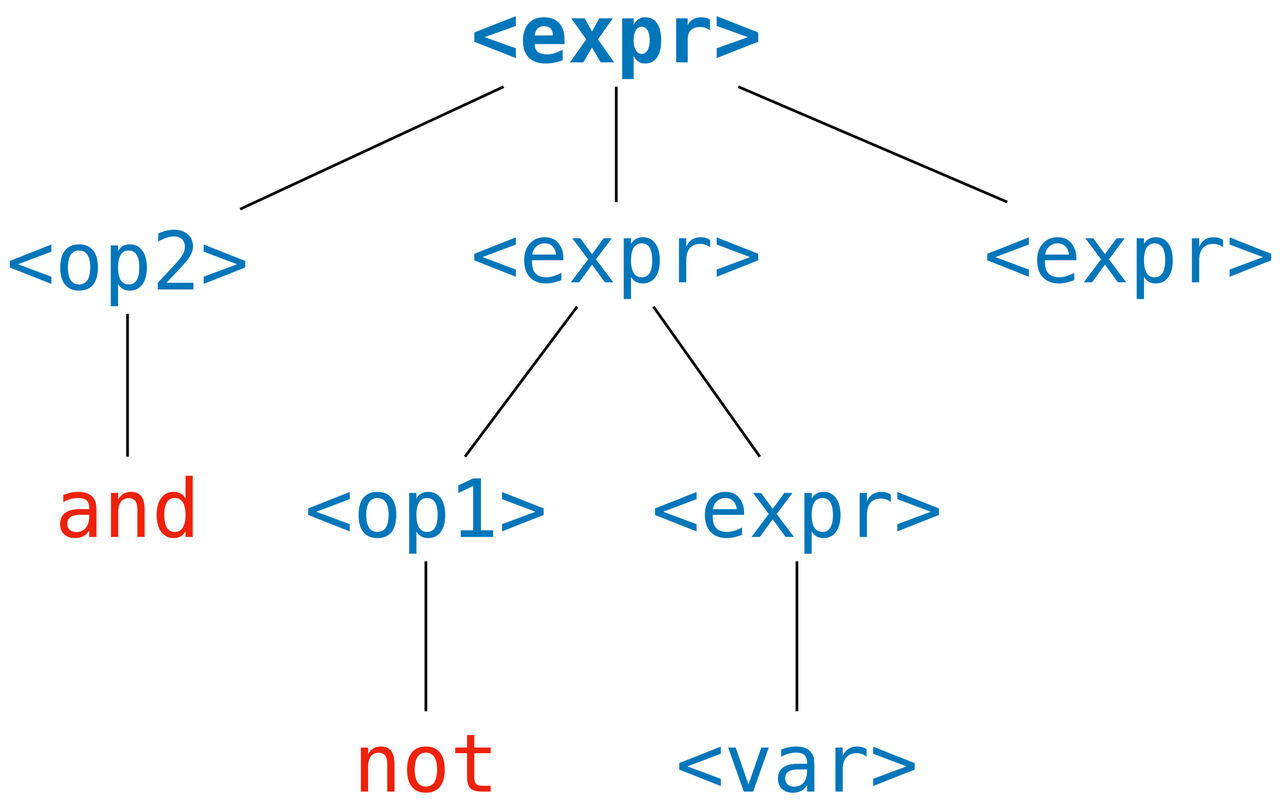

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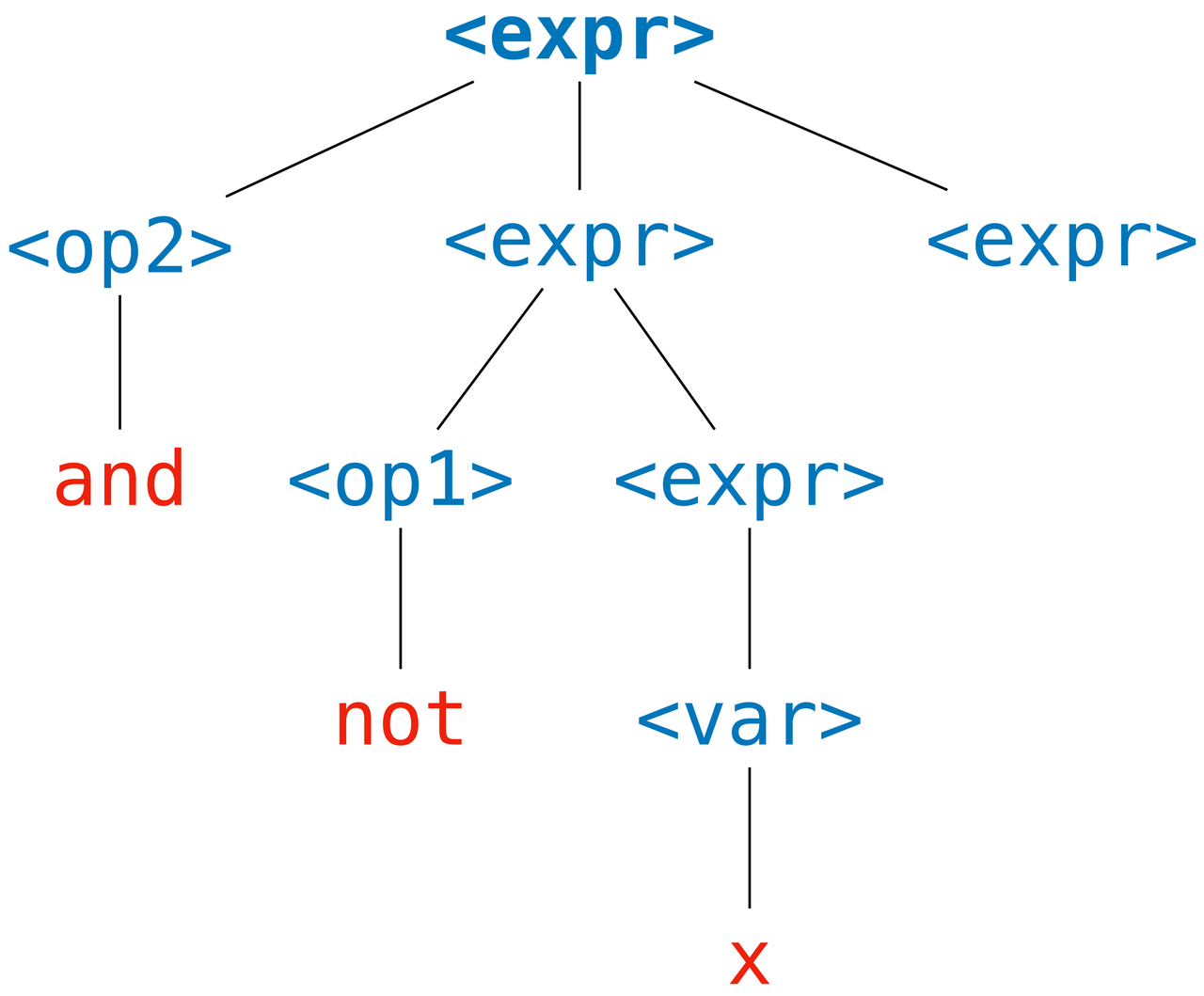

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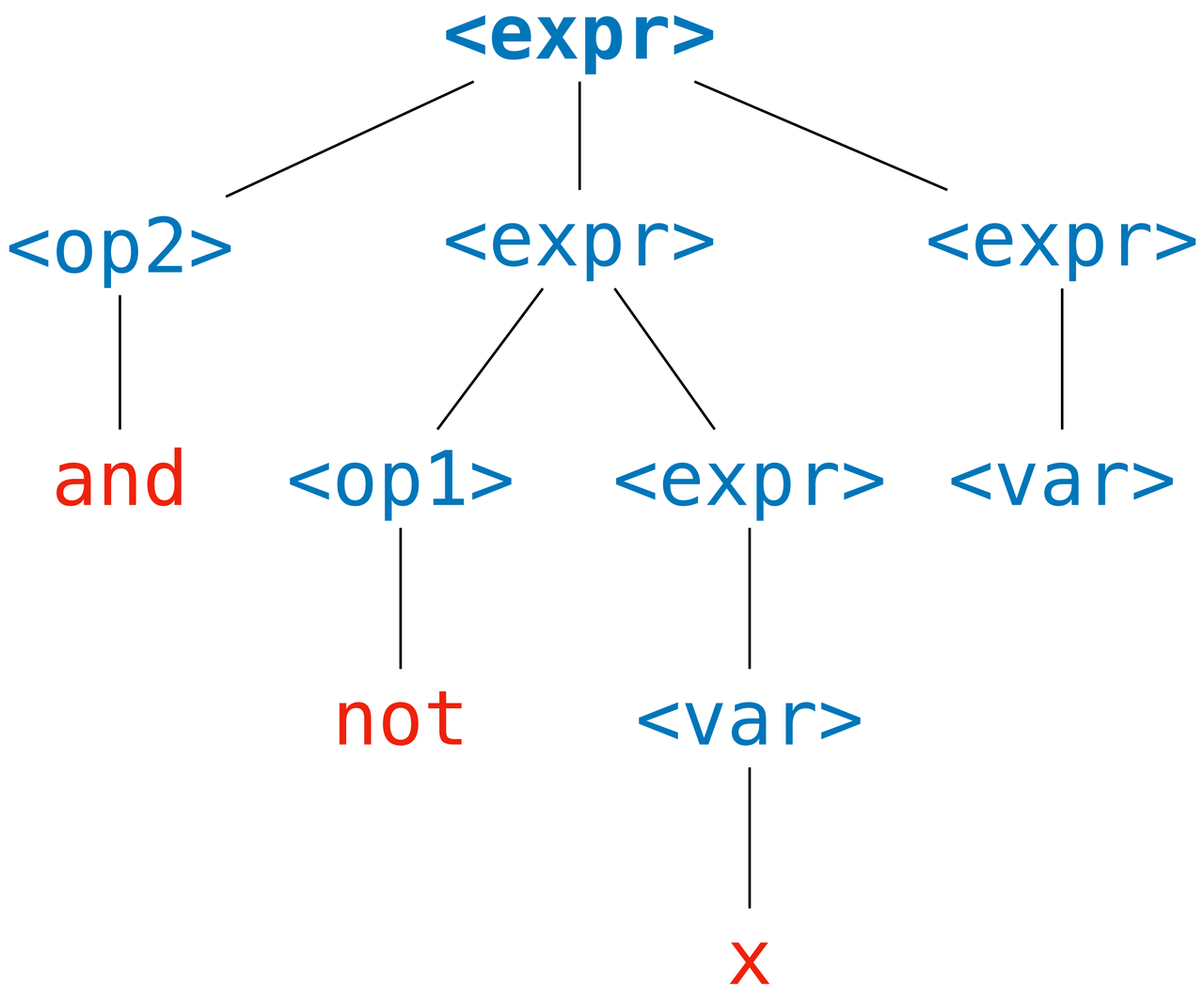

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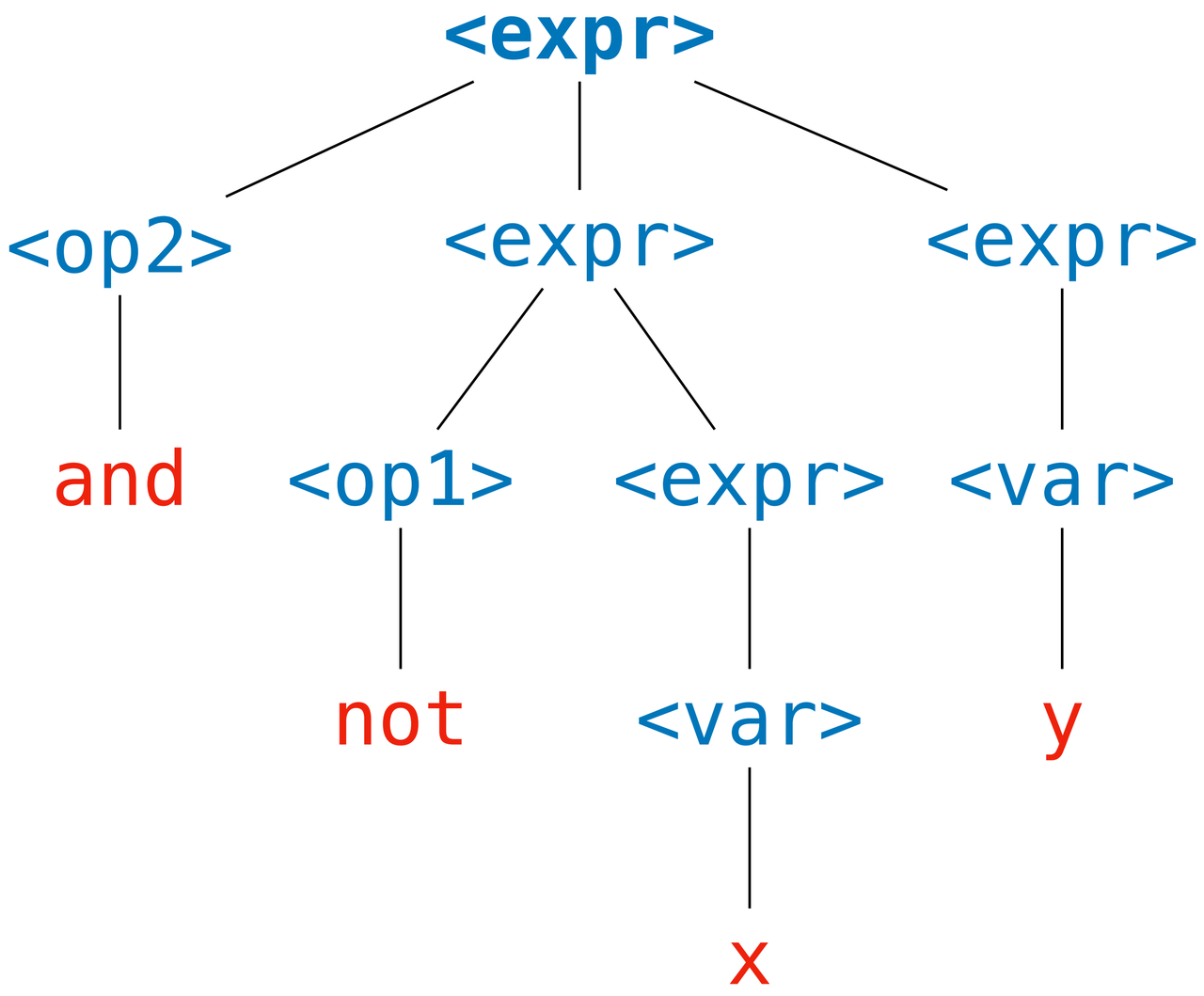

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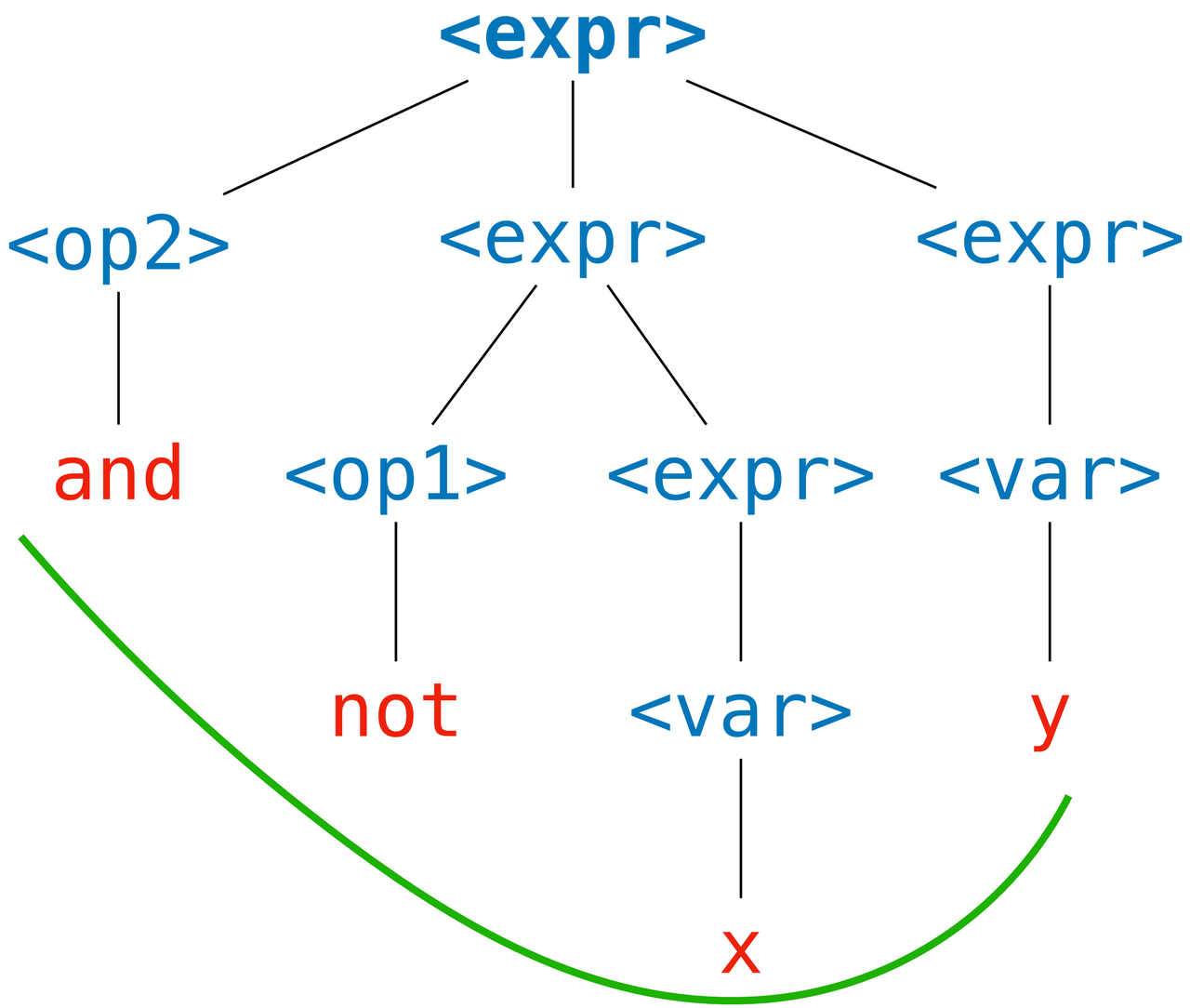

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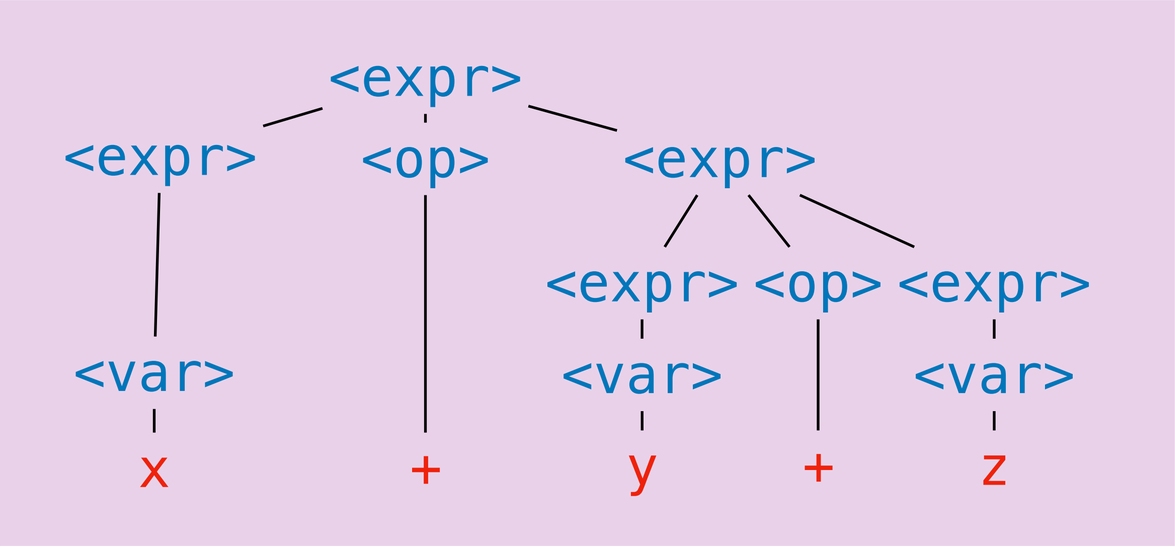

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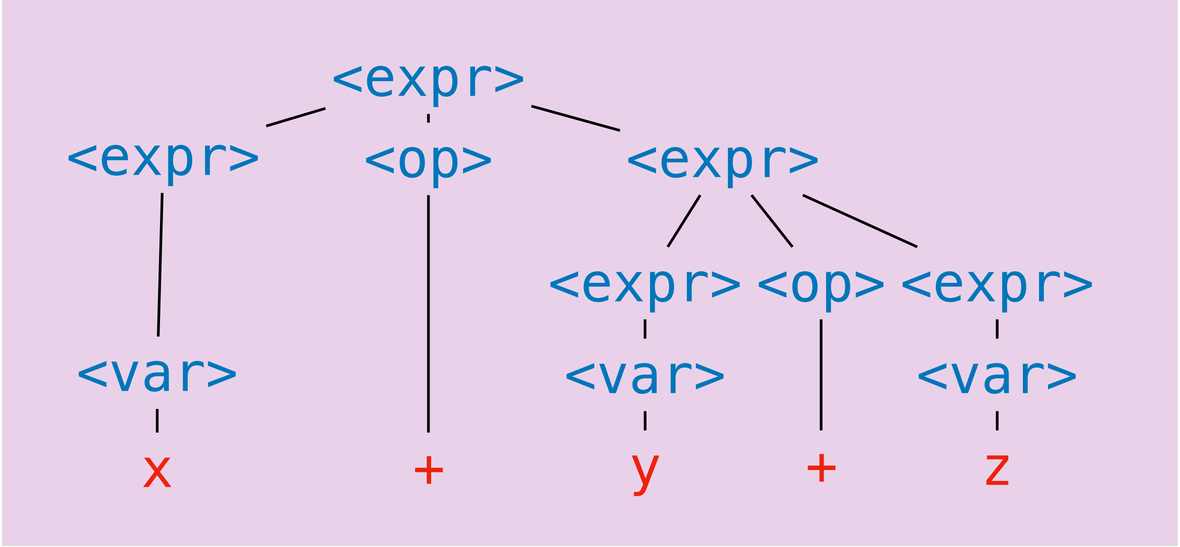

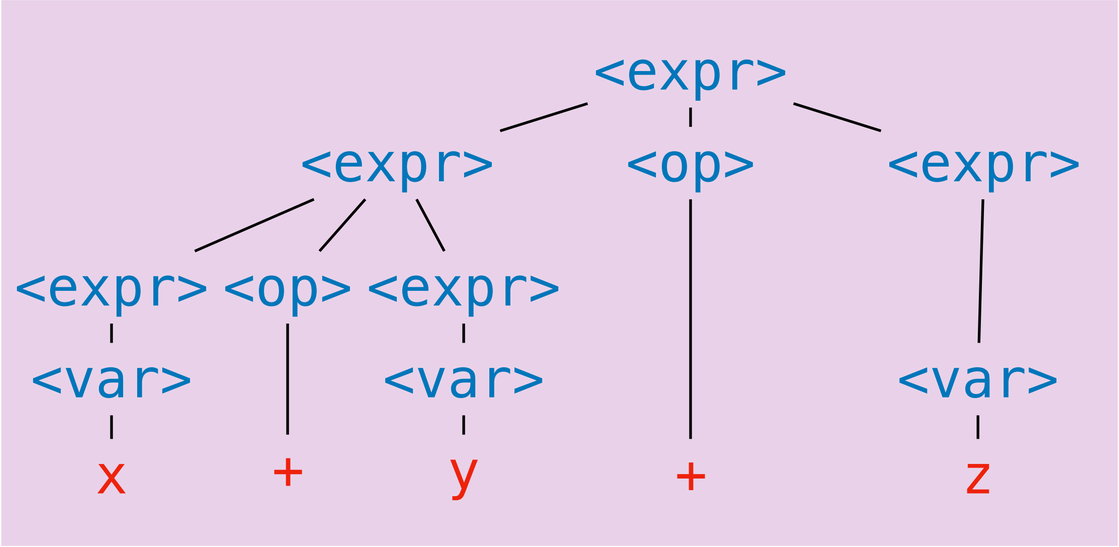

Practice Problem

```
<expr> ::= fun <var> -> <expr>
         | <expr> <expr>
         | <var>
<var>  ::= x
```

Demonstrate that the above grammar is ambiguous

Solution

```
<expr> ::= fun <var> -> <expr>
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<var>  ::= x
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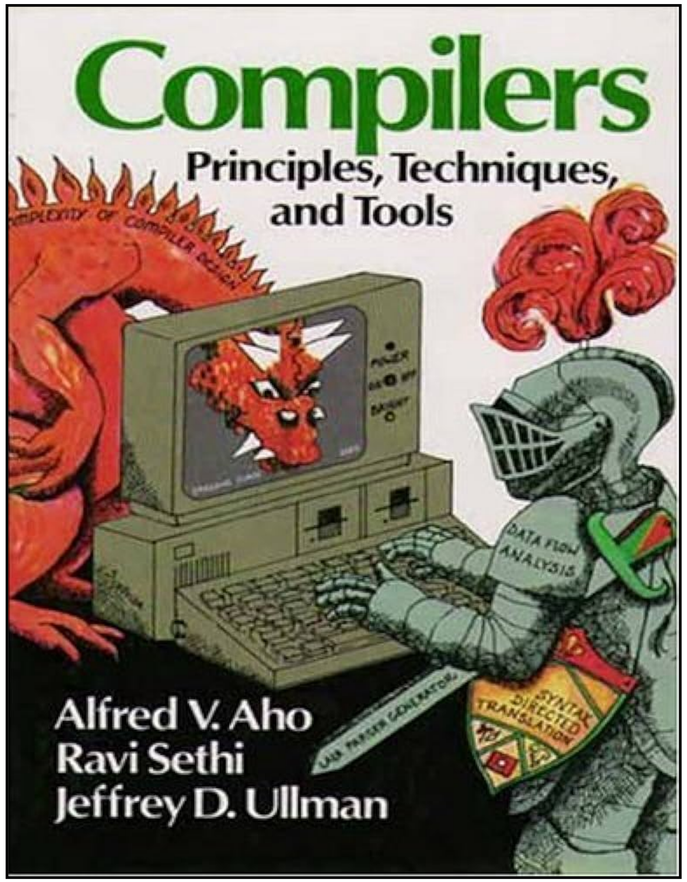
Motivation

A Note on "History"



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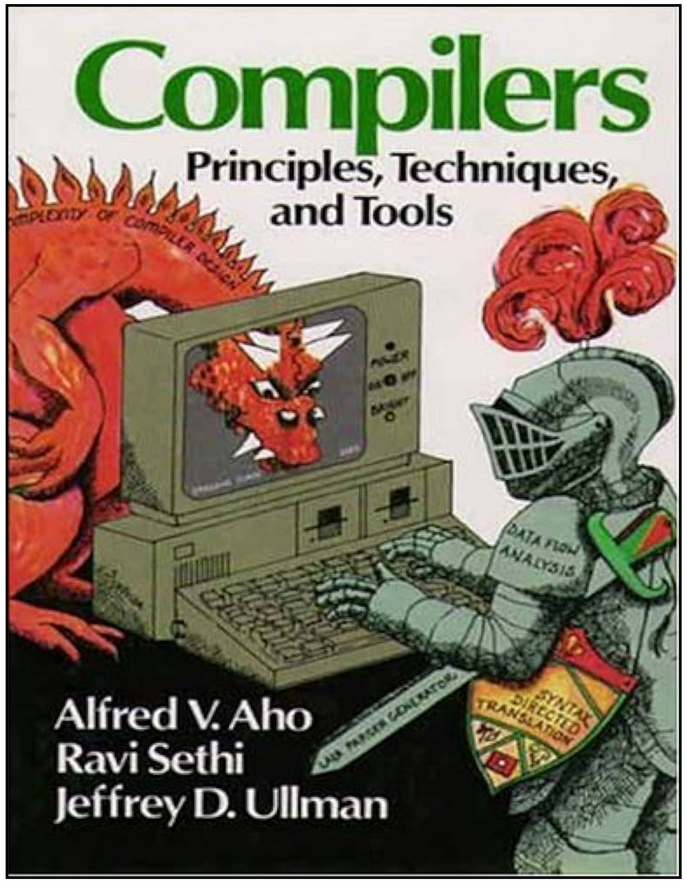
Lexing and parsing are typically part of ***compiler design***



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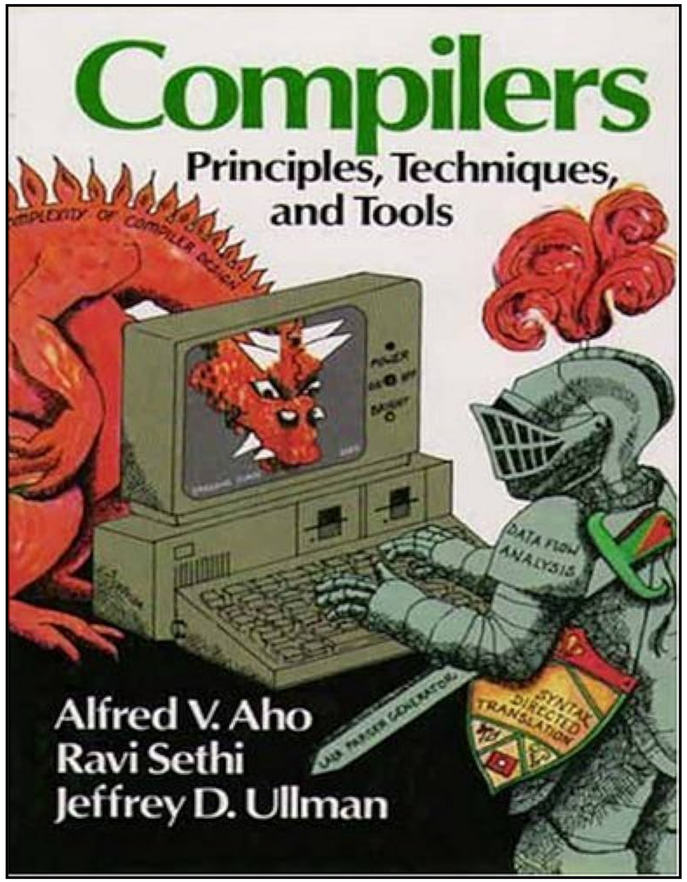

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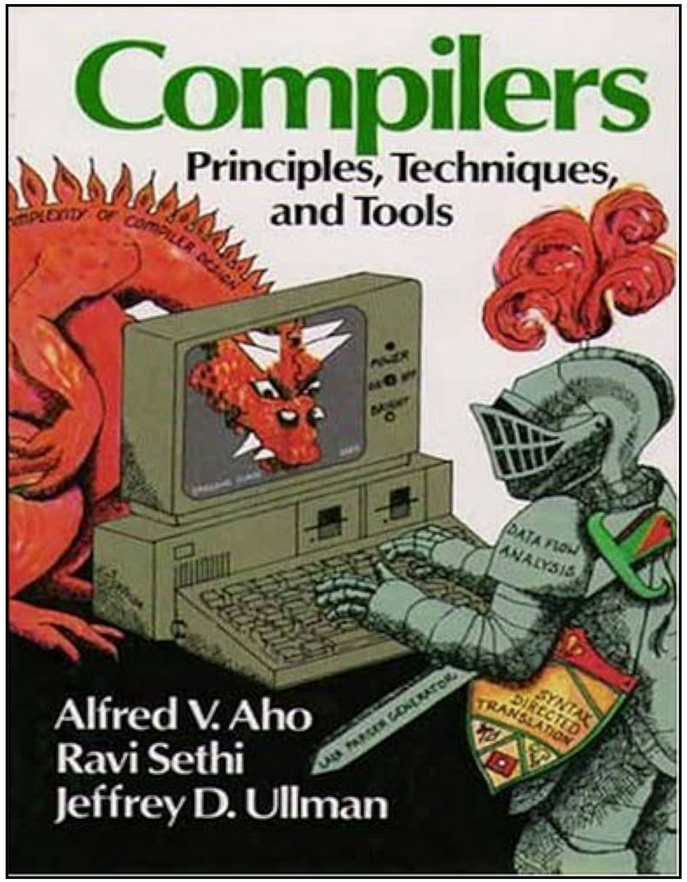

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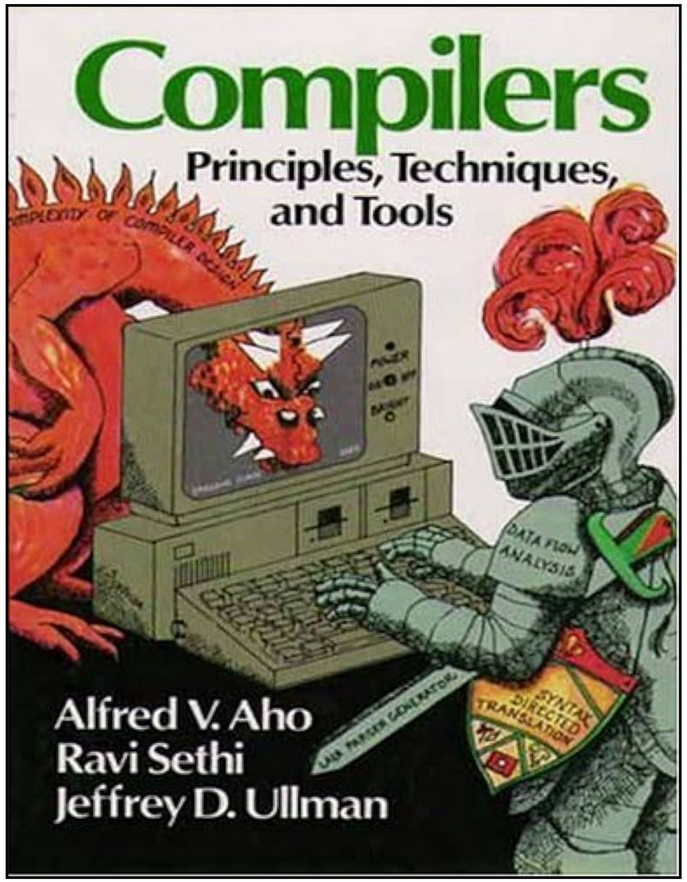
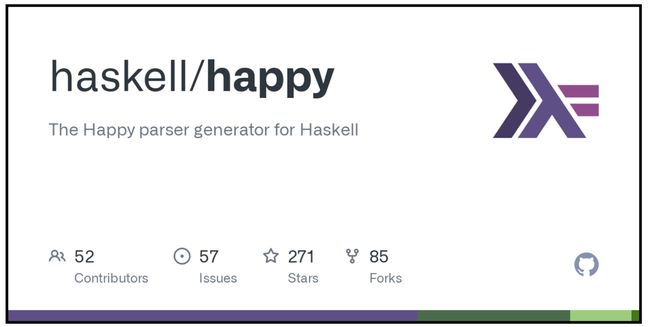
Also, we have ***parser generators***



Parser Generators



(Beaver)

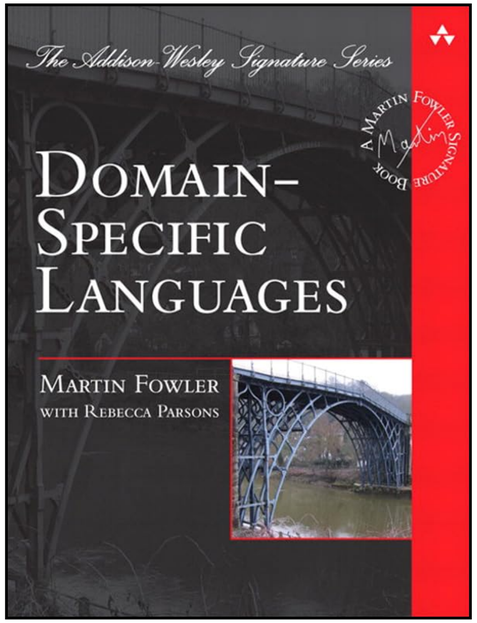
Parser generators are programs that build a parser given a *grammar*

Aside: Domain-Specific Languages

Domain-specific languages (DSLs) are programming languages for domain-specific tasks

» Emacs Lisp
» SQL

*We need **parsers** for these languages if we want to use them...*



SQL

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Note: EBNF means different things to different people

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Menhir:

```
expr =
  | IF; e1 = expr; THEN; e2 = expr; e3_opt = else?
    { match e3_opt with
      | None -> It (e1, e2)
      | Some e3 -> Ite (e1, e2, e3)
    }
else =
  | ELSE; e = expr { e }
```

Repetition Syntax

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Menhir:

```
elem =
  | SEMICOLON; e = expr { e }
expr =
  | LBRAK; RBRAK { List_lit [] }
  | LBRAK; e = expr; es = elem*; RBRAK
    { List_lit (e :: es) }
```

Interlude: Regular Expressions

Regular Grammars

```
<nonterminal> ::= terminal
<nonterminal> ::= terminal <nonterminal>
<nonterminal> ::= ϵ (the empty string)
```

A **regular grammar** is a BNF grammar with the above kinds of rules

Regular grammars are easier to parse

Example

```
<s> ::= a <s>
<s> ::= b <a>
<a> ::= ϵ
<a> ::= c <a>
```

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Example

```
<s> ::= a <s>
<s> ::= b <a>
<a> ::= ϵ
<a> ::= c <a>
```

is equivalent to

```
a*bc*
```

or

```
'a'* 'b' 'c'*
```

in ocamllex syntax

Example: Numbers and Variables

```
-?[0-9]+
```

numbers

```
[a-z_][a-z0-9A-Z_']*
```

variables

We'll leave it there, take CS332 if you want more, or read the Wikipedia page...

Lexical Analysis

The "Lexing" Problem

$$\texttt{"let"} \approx [\texttt{'l'}, \texttt{'e'}, \texttt{'t'}] \mapsto \textit{LET}$$

$$\texttt{"fun"} \approx [\texttt{'f'}, \texttt{'u'}, \texttt{'n'}] \mapsto \textit{FUN}$$

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» Characters are grouped into *smallest units* of the language

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» Characters are grouped into *smallest units* of the language

» Whitespace and comments are *ignored*

The "Lexing" Problem

$$\texttt{"let"} \approx \texttt{['l', 'e', 't']} \mapsto \textit{LET}$$

$$\texttt{"fun"} \approx \texttt{['f', 'u', 'n']} \mapsto \textit{FUN}$$

The Goal. *Convert a stream of chars to a stream of tokens*

- Characters are grouped into *smallest units* of the language
- Whitespace and comments are *ignored*
- Syntax errors are caught, when possible

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» expressions, statements, modules

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» **Simplicity.** It's *easier to think about* parsing if we don't need to worry about whitespace, characters, etc.

» **Portability.** Files & char encodings are finicky, handled differently across different operating systems. *Abstracting this away* is just good software engineering

Lexemes and Tokens

input program:	fun	l	->	l	++	[	100	]		
lexemes:	"fun"	"l"	"->"	"l"	"++"	"["	"100"	"]"		
tokens:	FUN	(ID "l")	ARR	(ID "l")	(OP "++")	LBRAK	(INT 100)	RBRAK		

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» "12" and "234" are INT_LITS, "let" is a KEYWORD

Parsing with Menhir

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In Theory. *Determine if a given sentence is recognized by a given grammar*

In Practice. *Given a grammar, write a program which converts a string recognized by that grammar into an ADT*

Today

```
<prog>  ::= <expr>

<expr>  ::= let <var> = <expr> in <expr>
          | <expr1>

<bop>   ::= + | - | * | /

<expr1> ::= <expr1> <bop> <expr1>
          | <num>
          | <var>
          | ( <expr> )

<num>   ::= 0 ; DUMMY VALUE
<var>   ::= x ; DUMMY VALUE

; In lex.mll:
;
; let num = '-'? ['0'-'9']+
; let var = ['a'-'z' '_'] ['a'-'z' 'A'-'Z' '0'-'9' '_' '\'']*
```

Operators in order of increasing precedence:

Operator	Associativity
+, -	left
*, /	left

We'll be building a parser for the this grammar

A Rough Sketch

1. Specify the tokens (i.e., terminal symbols) of the grammar

2. Specify the rules of the grammar (using a BNF-like syntax)

3. Specify the rules of the lexer (i.e., which strings go to which tokens)